CODING PYTHON WITH AI TOOLS IN THE REAL WORLD

AN INTERACTIVE LIVE-CODING SESSION

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🤓 ABOUT ME // CODE AND SLIDES

Johannes Schöck

- Studied molecular nano science, PhD in physics
- Self-taught coding and data science skills
- Started transition into data science in 2019
- Data Scientist at NKD since 2023

Find me on LinkedIn

Code and slides

🔗 https://github.com/JSchoeck/talk_agentic_coding_with_python

Key Facts and Figures

Leading platform in the attractive **discount sector**

~2,300 profitable, smaller format stores across 7 countries

+10,000 employees

850 Mio turnover in 2024

5% sales growth from 2023

Our Corporate Mission

NKD - A good feeling every day - for our customers, for our employees, for society. As a sympathetic discounter, we act responsibly, are always close by, and offer everyone an attractive value-for-money ratio.

European Footprint

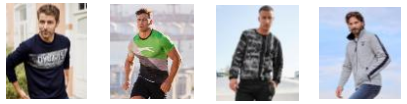
~2,300 stores in 7 countries

Category Overview

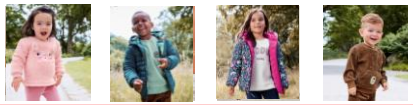
Apparel, +75%

Womenswear (64% of Apparel)

Menswear (19%)



Kidswear (17%)



Home, 15-20%

Homeware

Other, 5-10%

Other

WHAT WE WILL COVER TODAY

- What you need to get started with AI-assisted coding
- How AI fits into a real developer workflow
- Practical prompting techniques for Python
- How to structure an AI-assisted project to achieve the best results
- When to trust AI code – and when to double-check

🙋🏼‍♂️ WHO IN THE ROOM…?

Is coding regularly?

Uses AI-based coding assistants?

Plans to start using agents for coding?

📋 WHAT DO WE NEED?

- VS Code [1]
- Python [2]
- LLM model, optimized for coding tasks
- Agentic framework [3]
- GitHub account with activated Copilot

Why not bleeding edge tools?

[1] https://code.visualstudio.com/download
[2] https://marketplace.visualstudio.com/items?itemName=ms-python.python
[3] https://marketplace.visualstudio.com/items?itemName=GitHub.copilot-chat

💻 LET'S CODE!

Who wants to build something?

🤔 WHAT TO ASK?

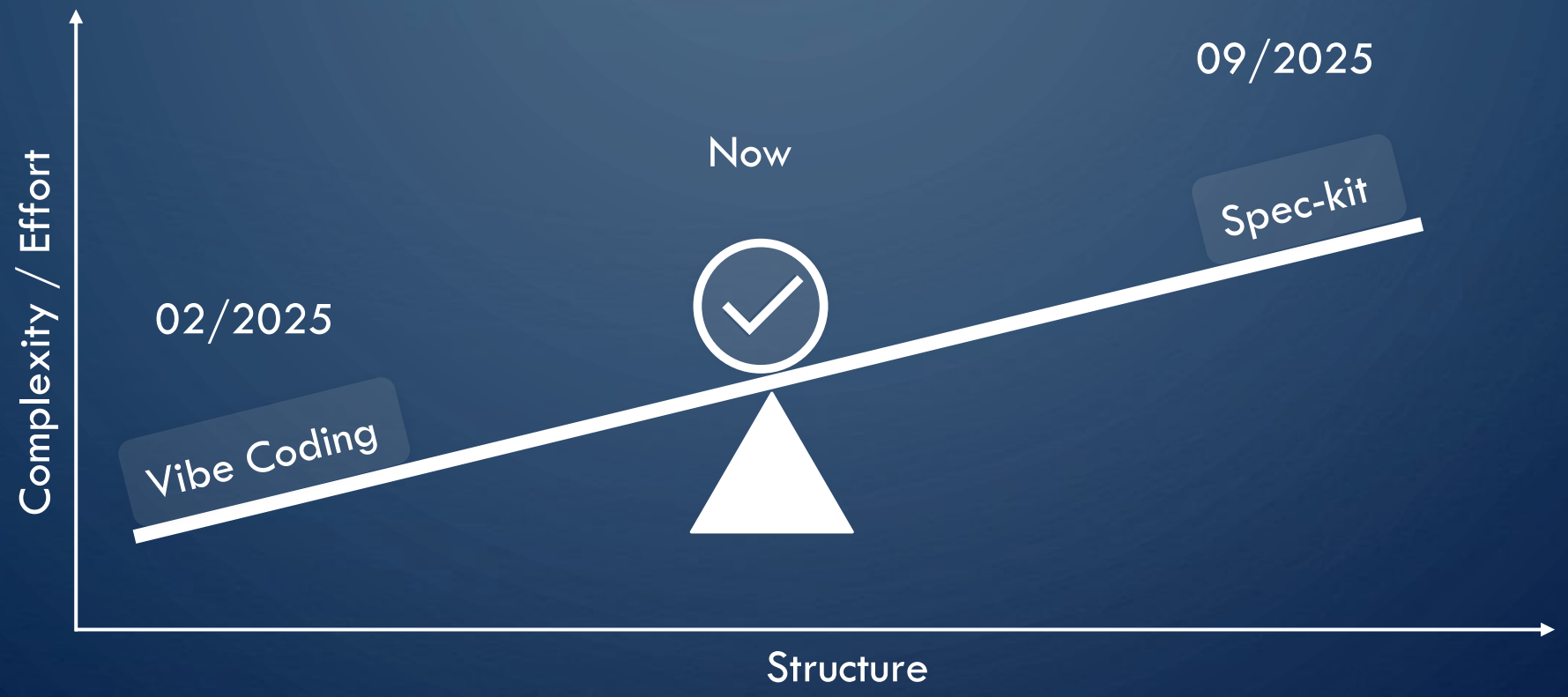
- Describe your goal
- Provide requirements, context and constraints
- Break down complex task into smaller parts
- Specify preferred technical choices
- Describe available data sources
- Add example input and output data format

➡ Provide sufficient context

WHAT TO DO WHILE WE WAIT?

- Think about
  - Business processes and their edge cases
  - User interface and experience
  - Input, processed, and output data
  - Interfaces to APIs
- Improve project description or documentation
- Review generated code more in depth
  - Check calculations and data processing steps
  - Tune layout / theme / user experience
  - Note next set of changes
- Improve generated code (comments, type hints, conventions)
- Prepare next iteration / feature

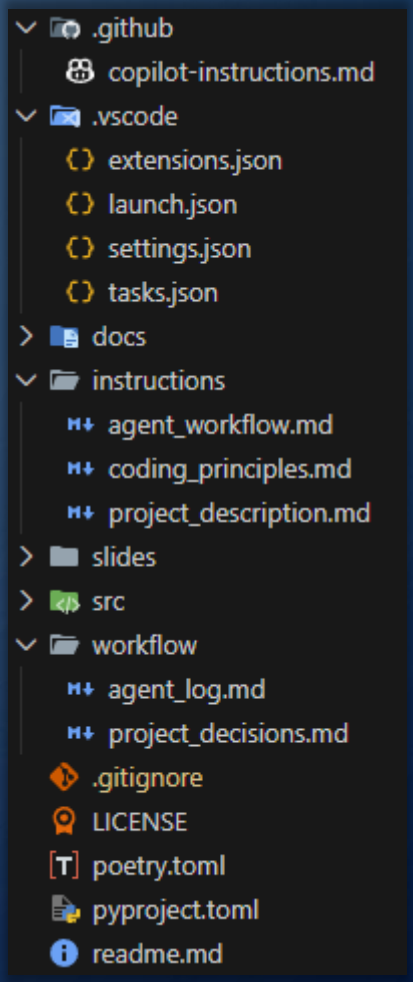
🛠️ SOME STRUCTURE CAN HELP



[1] https://github.com/github/spec-kit

🛠️ SOME STRUCTURE CAN HELP

- Use an instructions / agent files with useful guidelines [1]:
  - Tech stack decisions and external dependencies to use
  - How to use internal libraries and templates
  - Local system and deployment peculiarities
  - Business context
- Specify libraries and frameworks you're familiar with
- Add template files, if applicable, and mention them in context



[1] https://docs.github.com/en/copilot/how-tos/configure-custom-instructions/add-repository-instructions

⏩ WE GOT CODE, NOW WHAT?

- If the last iteration was not useful, go back one checkpoint
- Review code and decide what to keep, change, or undo
- If your current issue is completed, start a new chat session to not exceed context limit – avoid context compression as instructions get lost!
- Commit every useful addition into repository, even if only locally

WHAT TO LOOK OUT FOR

- Modified existing, human-written code, without asking. Make sure you agree with the changes and that they do not change expected behavior.
- Hallucinated functions, parameters, or even whole packages
- Granted code critical security permissions, e.g. *subprocess.run* with *shell=True*)
- Merged data on index, when indices do not actually line up correctly
- Added “guard clause” against missing data by returning empty data variable
- Hard-coded values, instead of using global definitions or configuration files
- Introduced excessive guard clauses, e.g. is a variable an *int*, if the function to assign it can only return *int*

📈 HOW TO IMPROVE RESULTS

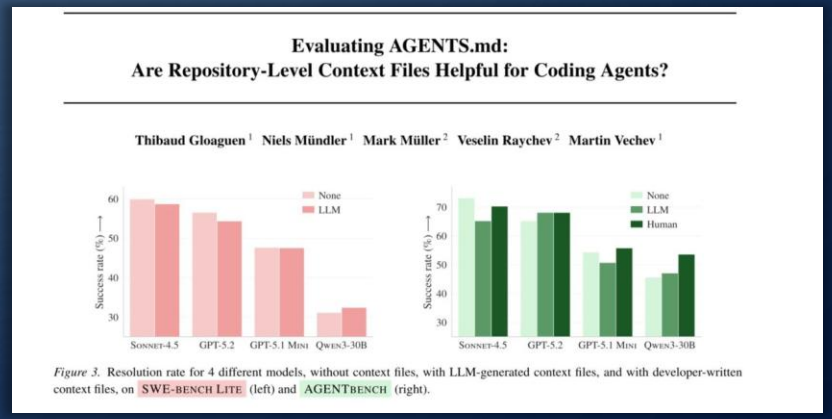
- Add more precise instructions what you do want, including examples, and what you do not want
- Do not fix bad output, instead diagnose, improve instructions, reset and re-run
- Use type hints, linter and formatter (e.g. ruff [1]) to spot issues

- Choose a different model (unlock new releases), especially when returning
- Build a complete agent workflow system (hooks, testing, multi-agent) [2]

https://docs.github.com/en/copilot/get-started/best-practices
[1] https://docs.astral.sh/ruff/
[2] https://youtu.be/88FC685v7ac?is=Mubw2hdHXXRIIXz0

⚠️ WARNINGS

- Skills and other instructions can be malicious [1]
- Agents who retrieve data from the web can be fed malicious instructions [2]
- Be very, very careful with auto-accept! Best: only use agents in a sandbox
- Quick but imprecise coding can lead to large technical debt in the future – make sure to enforce good code quality, or only use for prototyping
- Don't use autogenerated basic instructions, not worth it [3]



[1] "Be Careful w/ Skills", YouTube, The PrimeTim,e https://www.youtube.com/watch?v=Y2otN_NY75Y
[2] https://cybernews.com/security/hackers-poison-websites-with-malicious-ai-prompts/
[3] https://www.linkedin.com/posts/eric-vyacheslav-156273169_researchers-just-proved-that-agentsmd-files-share-7431956174654226432-8pbY

📖 WHERE TO LEARN MORE

- https://drivendata.co/blog/ai-agents-data-science-competitions
- https://code.visualstudio.com/docs/copilot/overview
- Podcasts, YouTube & GitHub
  - Folder System for agentic coding [1]
  - Episode #277: Moving Towards Spec-Driven Development – The Real Python Podcast [2]
  - Local coding agents in 2026 [3]
  - GitHub spec-kit & orchestration with AI toolkit [4]
  - karpathy/autoresearch: AI agents running research, training automatically

[1] https://www.youtube.com/watch?v=MkN-ss2NI10
[2] https://realpython.com/podcasts/rpp/277/
[3] https://youtu.be/3zSANOIBHYw
[4] https://www.youtube.com/watch?v=ob-TOPfQmbk

- https://docs.github.com/de/copilot
- https://docs.github.com/de/copilot/reference/customization-cheat-sheet
- https://vibeskills.eu by Benjamin Linnik

QUESTIONS

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